KAVIBHARATHI BALAKRISHNAN

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PROFESSIONAL OBJECTIVE

Agricultural Engineering graduate with hands-on expertise in GIS, Remote Sensing, and CAD-based design tools, seeking an entry-level role in precision agriculture, geospatial analysis, or agri-tech product development. Skilled in satellite image processing, multispectral drone data, spatial mapping using QGIS/ArcGIS, 2D drafting in AutoCAD, and 3D mechanical design in SolidWorks — committed to delivering data-driven, field-ready solutions for sustainable agriculture.

SKILLS

GIS & Geospatial: QGIS (Basic to Advanced), ArcGIS

Cartography: Map Layout & Output

Design & CAD Tools: AutoCAD, SolidWorks

Other: Video Editing

Languages: Tamil, English

Agriculture Domain: Soil and water conservation, Farm machinery, Renewable energy, Remote sensing and GIS

INTERNSHIP EXPERIENCE

Agricultural Machinery Trainee — CAI Auto Industries Pvt. Ltd. *July 2024*

- Studied design and operational mechanics of Mahindra tractor models and implement attachments used in field operations.
- Gained insight into tractor PTO systems, hydraulic linkages, and implement compatibility standards.
- Applied AutoCAD drafting knowledge to understand mechanical assembly drawings and component specifications.

Agri-Business & Production Intern — Elies Biotech Pvt. Ltd. *December 2024*

- Contributed to marketing strategy and project management for organic fertilizer and bio-input product lines.
- Studied manufacturing of compost-based, bio-stimulant, and microbial fertilizers for field application.
- Strengthened cross-functional communication and team collaboration in an agri-startup environment.

GIS & Remote Sensing Intern — Alpha Map Tech

- Executed geospatial workflows in QGIS (Basic to Advanced): vector/raster analysis, spatial joins, and attribute management.
- Used ArcGIS Pro for buffer analysis, overlay operations, geoprocessing, and cartographic map production.
- Integrated drone-captured imagery with GIS platforms for land-use mapping and agricultural field delineation.
- Produced georeferenced maps and spatial outputs aligned with real-world remote sensing workflows.

KEY PROJECTS

AI-Powered Harvester for Vertical Farming — Prototype Development

- Designing an AI-driven robotic harvester with multispectral sensors for real-time crop maturity detection in vertical farms.
- Developing mechanical chassis and actuator systems using SolidWorks 3D modelling; structural components drafted in AutoCAD.

Crop Stress Detection System — Final Year Project

- Deployed UAV drones with multispectral cameras to capture high-resolution aerial imagery of agricultural plots.
- Integrated raster layers with QGIS/ArcGIS for spatially explicit stress-pattern maps enabling targeted agronomic intervention, combined with artificial intelligence.
- Outputs support precision input application, reducing chemical overuse and improving crop yield forecasting accuracy.

PRESENTATIONS & EVENT PARTICIPATION

- Presented a paper at a National-level Technical Symposium on precision agriculture and geospatial applications in modern farming.
- Attended a Remote Sensing & GIS Workshop: satellite data interpretation, band compositing, and thematic map creation.
- Participated in a Survey & Levelling Workshop: total station operation, GPS field data collection, and topographic map generation.

EDUCATION

B.Tech, Agricultural Engineering — Sri Shanmugha College of Engineering & Technology

CGPA: 8.5 | 2022 – 2026

Higher Secondary (HSC) — Govt. HSS Devanankurichi

Score: 8.5 | 2021 – 2022

SSLC

Score: 9.0 | 2019 – 2020

DECLARATION

I hereby declare that all the information furnished above is true and correct to the best of my knowledge and belief. I take full responsibility for the authenticity of the details provided in this resume and I understand that any misrepresentation may disqualify my candidature.